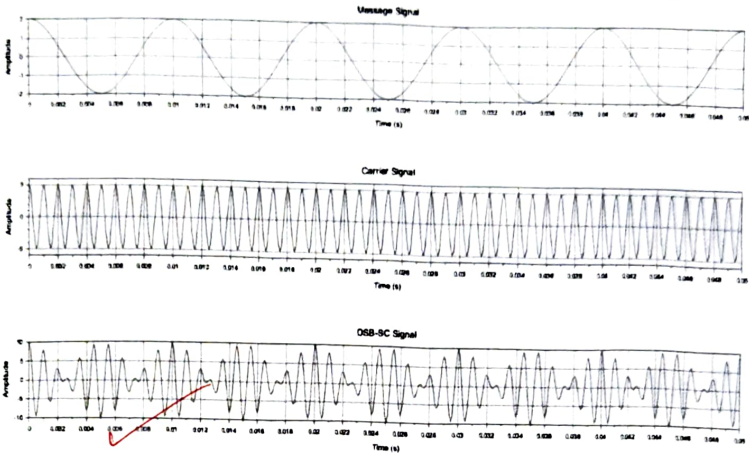


Scilab window number 1
File Tools Edit
Graphic window number 1



opp

scilab program & Implementation (20)	Waveform & calculation (20)	calculation- and result (20)	viva-voce (20)	Result & observat ⁿ (20)
20	20	20	15	$\frac{20}{75}$

25/7/26

EXPERIMENT 2

DOUBLE SIDEBAND SUPPRESSED CARRIER

AIM:-

To generate and analyse of double-sided band suppressed carrier (DSB-SC) modulation.

EQUIPMENT REQUIRED:-

Scilab software

CODE:-

```
clc;
clear;
close;
t = 0:0.0001:0.05;
Am = 2;
Ac = 5;
fm = 10;
fc = 1000;
t = 0:1/fs:0.05;
m = Am * cos(2 * %pi * fm * t);
c = Ac * cos(2 * %pi * fc * t);
dsb = m * c;
subplot(3,1,1);
plot(t, m);
title("Message signal");
```

S.No	Particular	Theory		Practical		
		Amplitude	Frequency	Amplitude	Time	Frequency
1.	Message signal	2	100	2	$T = x_2 - x_1$ $\Rightarrow 0.020 - 0.010$ $= 0.01s$	$f_m = \frac{1}{T_m}$ $= \frac{1}{0.01} \Rightarrow 100Hz$
2.	carrier signal	5	1000	5	$T = x_2 - x_1$ $= 0.002 - 0.001$ $= 0.001s$	$f_c = \frac{1}{T_c}$ $= \frac{1}{0.001}$ $= 1000Hz$
3.	USB	—	1100	10	$T = x_2 - x_1$ $= 0.00191 - 0.00100$ $= 0.00091s$	$f = \frac{1}{T}$ $= \frac{1}{0.00091}$ $= 1100Hz$
4.	LSB	—	900	10	$T = x_2 - x_1$ $\Rightarrow 0.00211 - 0.00100$ $= 0.00111s$	$f = \frac{1}{T}$ $= \frac{1}{0.00111}$ $= 900Hz$

```
subplot (3, 1, 2);
```

```
plot (t, c);
```

```
title ("Carrier signal");
```

```
subplot (3, 1, 3);
```

```
plot (t, dsb);
```

```
title ("DSB-SC signal");
```

CALCULATION:-

$$m_a(\text{theory}) = \frac{E_m}{E_c}$$

$$\Rightarrow \frac{2}{5} \Rightarrow 0.4$$

$$m_a(\text{practical}) = \frac{E_{\max} - E_{\min}}{E_{\max} + E_{\min}} \Rightarrow \frac{2}{5} \Rightarrow 0.4$$

$$\Rightarrow \cancel{m_a(\text{theory})} = m_a(\text{practical})$$

RESULT:-

DSB-SC is generated, USB & LSB are stimulate
thus ~~the~~ modulated index is ~~g~~ calculated.